

Auteur: Rob Willemsen
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$$z^4 + 4 = z^4 + 4z^2 + 4 - 4z^2$$

$$z^4 + 4 = (z^2 + 2)^2 - (2z)^2$$

$$z^4 + 4 = (z^2 - 2z + 2)(z^2 + 2z + 2)$$

$$z^2 - 2z + 2 = 0 \Rightarrow z = \frac{2+2i}{2} = 1+i \text{ of } z = \frac{2-2i}{2} = 1-i$$

$$z^2 + 2z + 2 = 0 \Rightarrow z = \frac{-2+2i}{2} = -1+i \text{ of } z = \frac{-2-2i}{2} = -1-i$$

Antwoord: De nulpunten in $\mathbb{C}$ zijn $1+i$, $1-i$, $-1+i$ en $-1-i$. De ontbinding over $\mathbb{R}[z]$ is $A(z) = (z^2 - 2z + 2)(z^2 + 2z + 2)$.

References